



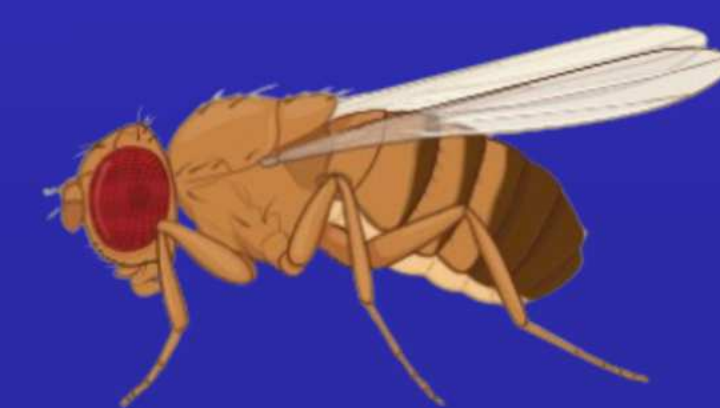
Figure 1: Oxidative Stress Expression (Made by Researcher in Biorender)

Targeting Stress-Induced Oxidative Damage

The Effect of *Zingiber Officinale* on Lipid Peroxidation-Associated Oxidative Stress through Malondialdehyde levels in Group-Housed and Socially Isolated *Drosophila melanogaster*

Drosophila melanogaster

"Fruit Fly"



- Genetically Relevant
- Cost-Effective
- Reproducible

Figure 14: D. Melanogaster (Made by Researcher in Biorender)

Introduction

- **Oxidative stress (OS)** occurs when **reactive oxygen species** production **exceeds antioxidant** defenses.
- ROS-driven **lipid peroxidation** damages **cell membranes** and generates **malondialdehyde (MDA)** (Biomarker for OS).

Disease	% of Cases Associated with OS
Cardiovascular Disease	100%
Parkinson's Disease	100%
Alzheimer's Disease	95%
Gastric Cancer	95%
Atherosclerosis	95%
Diabetes Mellitus	90%
Multiple Sclerosis	90%
Carcinogenesis	90%
Chronic Kidney Disease	85%
COPD	85%

Figure 2: OS Disease Association Table (Made by Researcher in Matplotlib)

- **TBARS** quantifies lipid peroxidation by **measuring MDA** as a concrete, numeric biomarker.
- Social stressors such as **social isolation** shift redox balance toward **higher ROS** and **higher MDA**.

Scientific Gap

- PROBLEM**

 - Current 3 therapeutics manage symptoms, not oxidative biomarkers: **LITHIUM CARBONATE, CLOZAPINE, OLANZAPINE.**
 - Harsh drawbacks: Serum monitoring, Thyroid and kidney risk, Severe neutropenia risk, Drives metabolic burden
- GAP**

No Intervention has been **validated** with a clean, controlled **dose-response** showing reduced oxidative biomarker levels.
- SOLUTION**

 - Identify a **novel therapeutic** candidate with both **Antioxidant** and **Neuroinflammatory** modulating properties.
 - Needs evidence of **lowering oxidative stress biomarkers.**

Hypothesis

- Statement**

 - **Ginger Essential Oil (GEO)**, derived from *Zingiber Officinale*, **fed flies** will show **lower MDA** than vehicle fed flies within each housing condition.
 - **Social isolation** will produce **higher MDA**, reflecting higher oxidative stress burden.
- Justification**

 - Prior Literature shows **ginger** extract **lowered MDA** in stressed rat cardiac homogenate tissue (Mao et al., 2019).
 - GEO contains strong & dominant **sesquiterpenes** such as α -zingiberene and 6-shogaol, with documented **antioxidant potential**.
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Figure 3: Oxidative Eustress Hypothesis Visual (Made by Researcher in Biorender)

Methodology

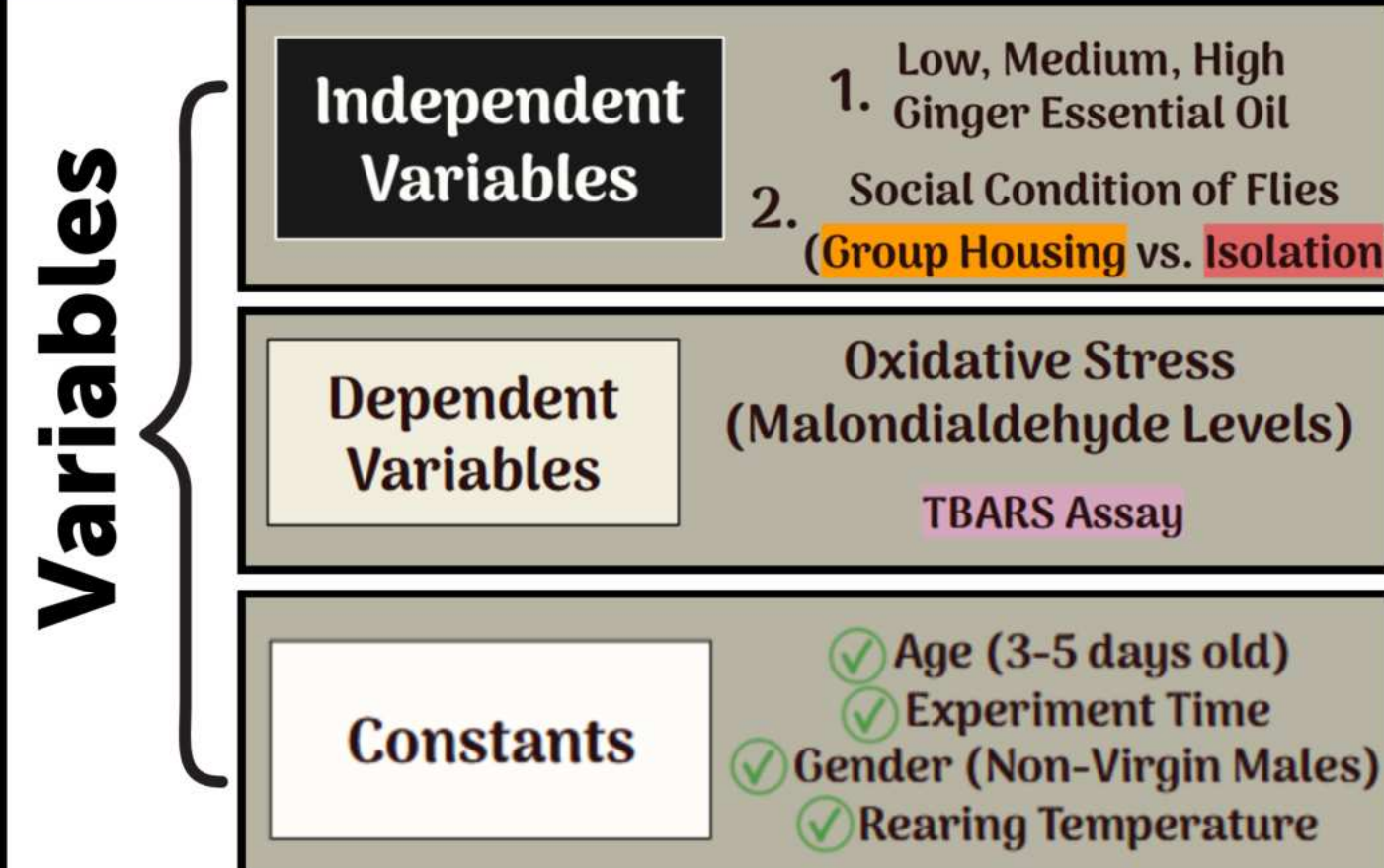


Figure 4: Variables Summary Visual (Made by Researcher in Google Slides)

Fly Group Factorial Design

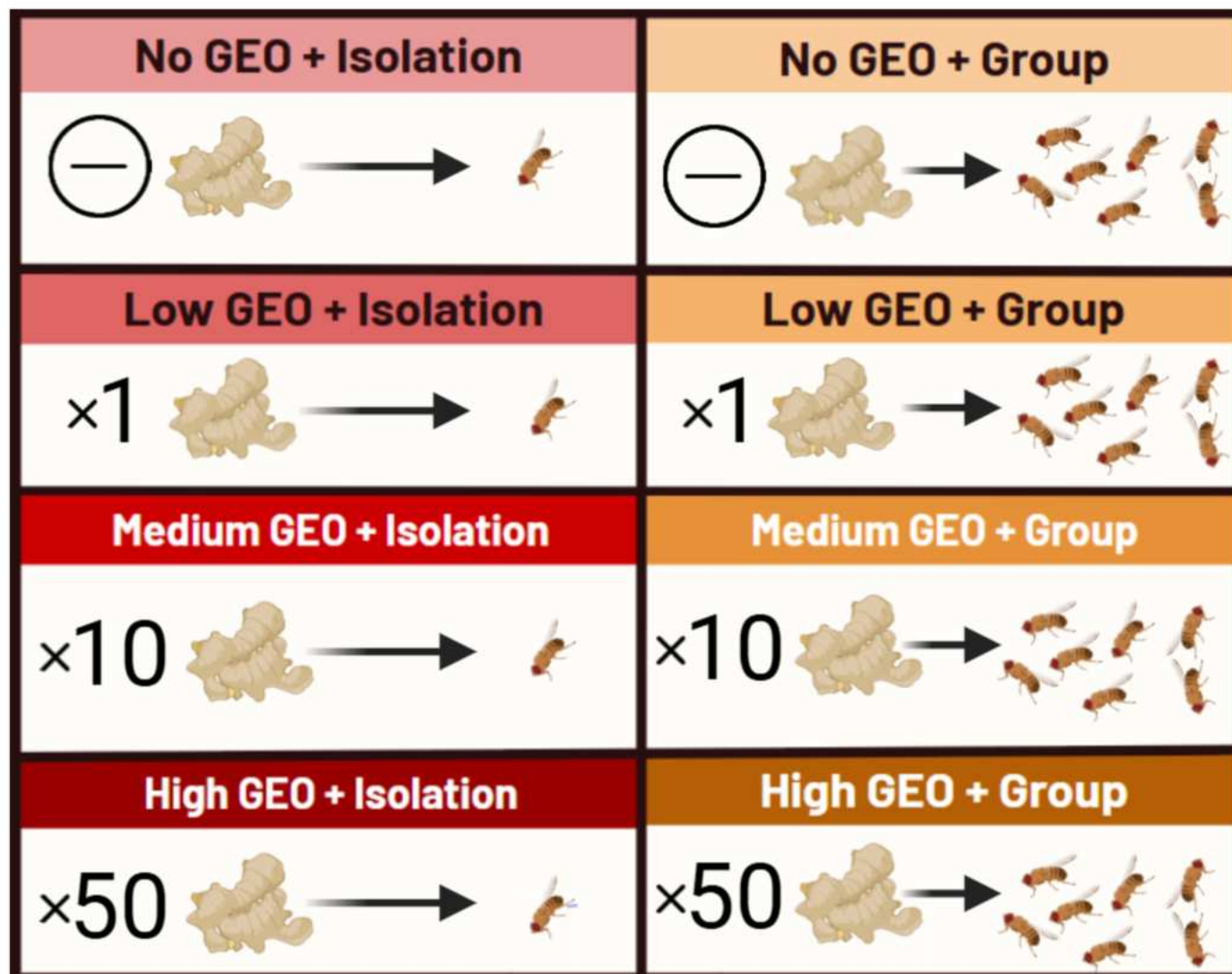


Figure 5: Eight Fly Groups (Made by Researcher in Biorender)

CO2 Anesthesia, Sorting, & Housing

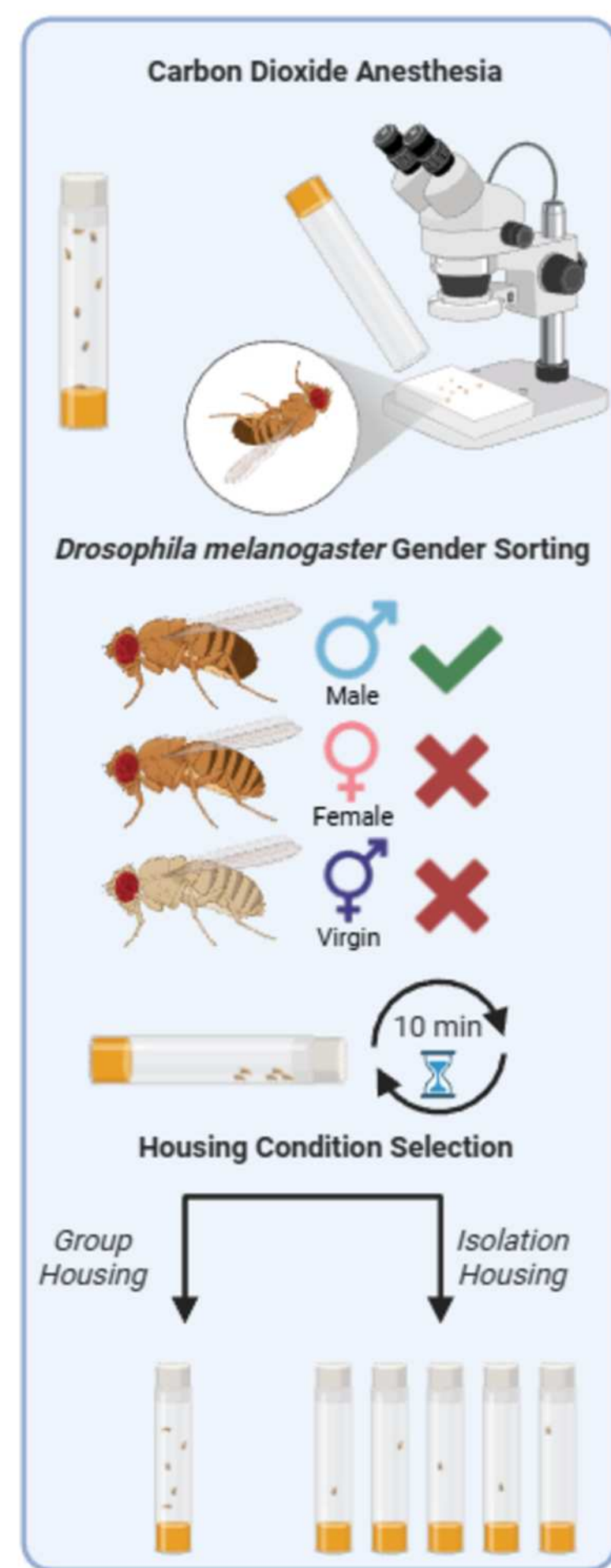


Figure 6: Anesthesia, Sorting, and Housing (Made by Researcher in Biorender)

Thiobarbituric Acid Reactive Substances Assay (TBARS)

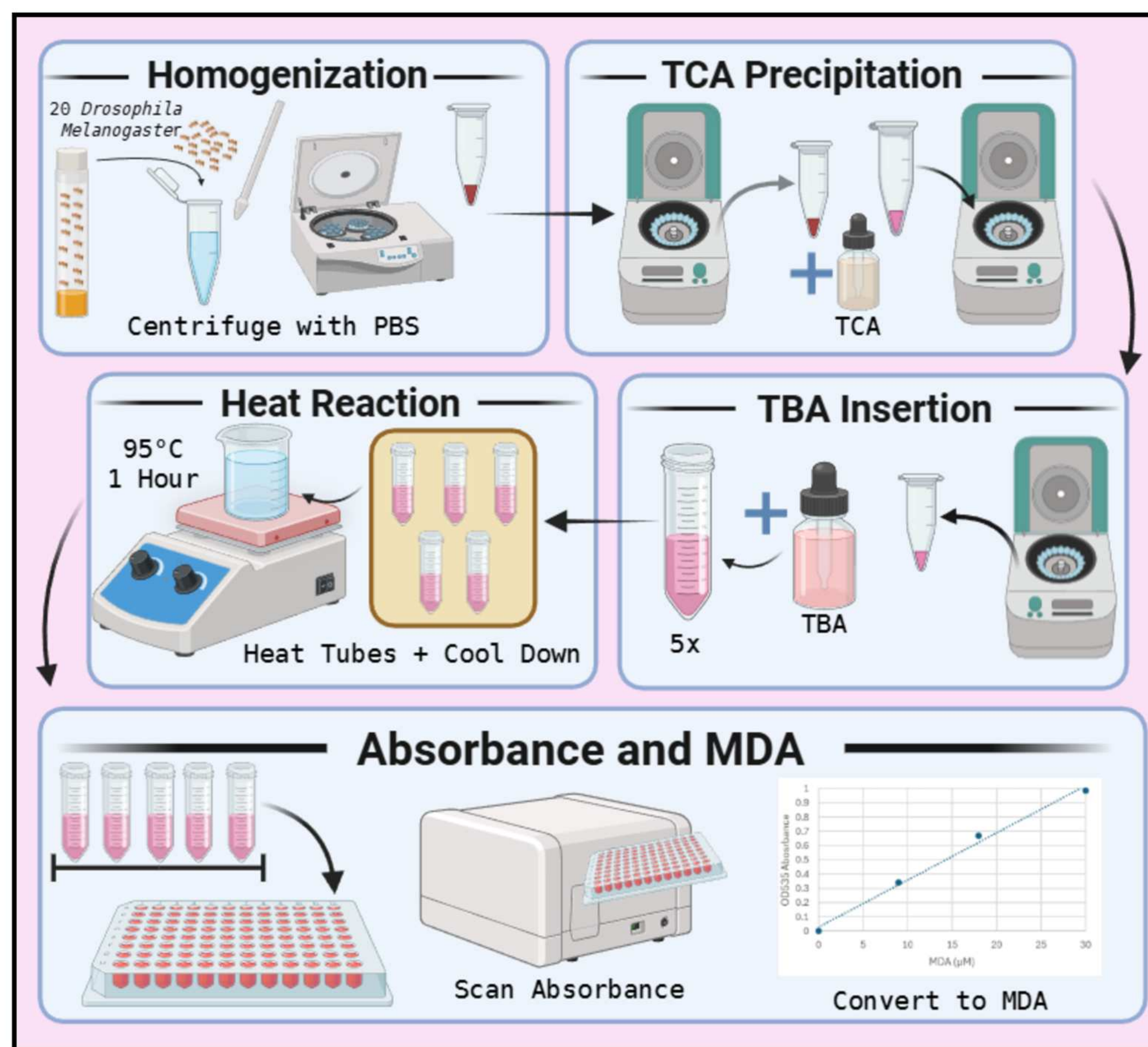


Figure 7: Thiobarbituric Acid Reactive Substances Assay Workflow (Made by Researcher in Biorender)

Method Novelty

Novel MDA quantification
Converted TBARS absorbance into **absolute MDA** (μM) using a standard curve, **unlike previous studies** that report raw OD or relative change without calibration.

Novel factorial design
Experimented with a **Housing x Dose** framework with eight conditions, separating stress effects from treatment effects in one controlled experiment.

Results & Findings

TBARS MDA Standard Results

No	30 μM MDA + H ₂ O	Vol (μL)	MDA (μM)
1	300 μL + 0 μL	300	30.0
2	180 μL + 120 μL	300	18.0
3	90 μL + 210 μL	300	9.0
4	0 μL + 300 μL	300	0.0

Figure 8: MDA Standard Volume Ratios for Curve (Made by Researcher in Excel)

$$[\text{TBARS}] = \frac{R_{\text{Sample}} - R_{\text{Blank}}}{\text{Slope } (\mu\text{M}^{-1})}$$

Figure 9: TBARS MDA Equation (Made by Researcher in Google Docs)

$$A = \epsilon \cdot c \cdot l$$

absorbance molar path length absorptivity

Figure 10: Beer Lambert Law (Made by Researcher in Google Docs)

- Standards covered 0 to 30 μM MDA, producing absorbance from 0.000 to 0.987, giving a **wide calibration range** for sample conversion.
- Standard curve fit: Absorbance = 0.0330 $\times$ (MDA μM) + 0.0287 with $R^2 = 0.992$, **supporting strong linear quantification** across the standard range.

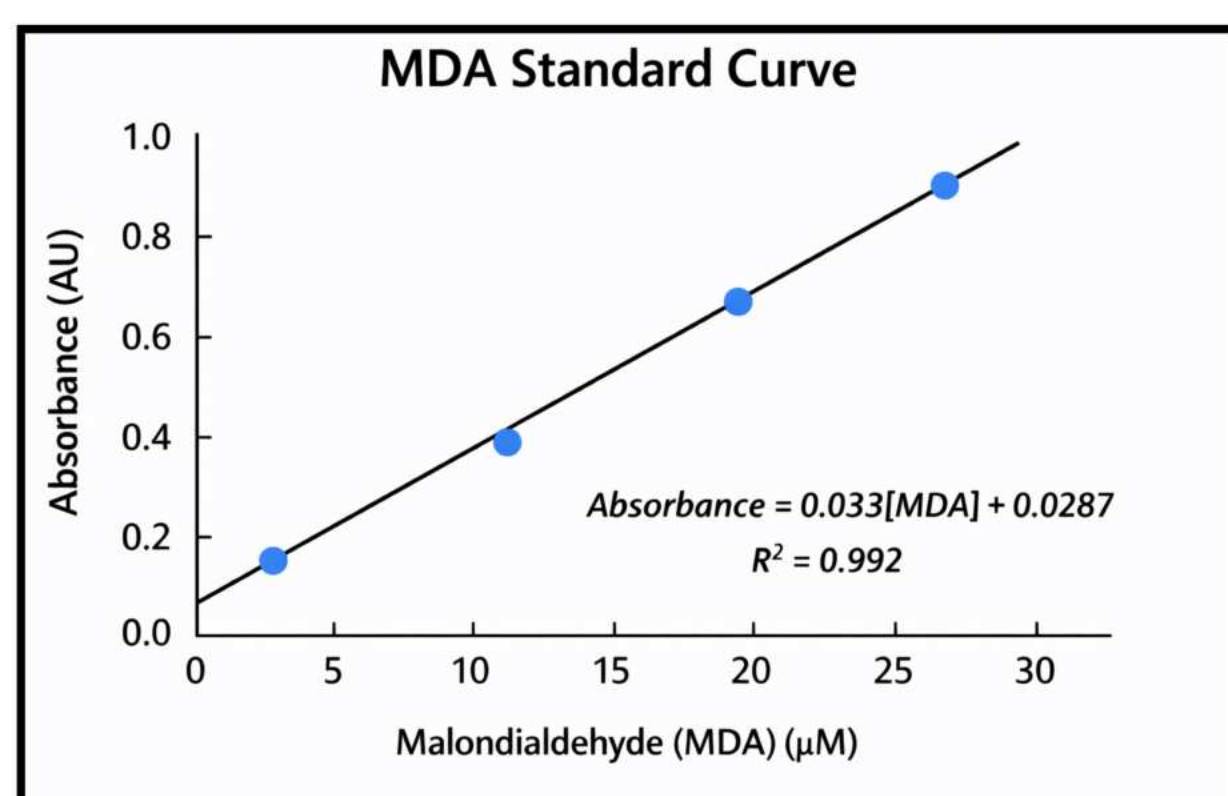


Figure 11: Malondialdehyde (MDA) Standard Curve (Made by Researcher in Matplotlib)

TBARS Isolation Experimental Results

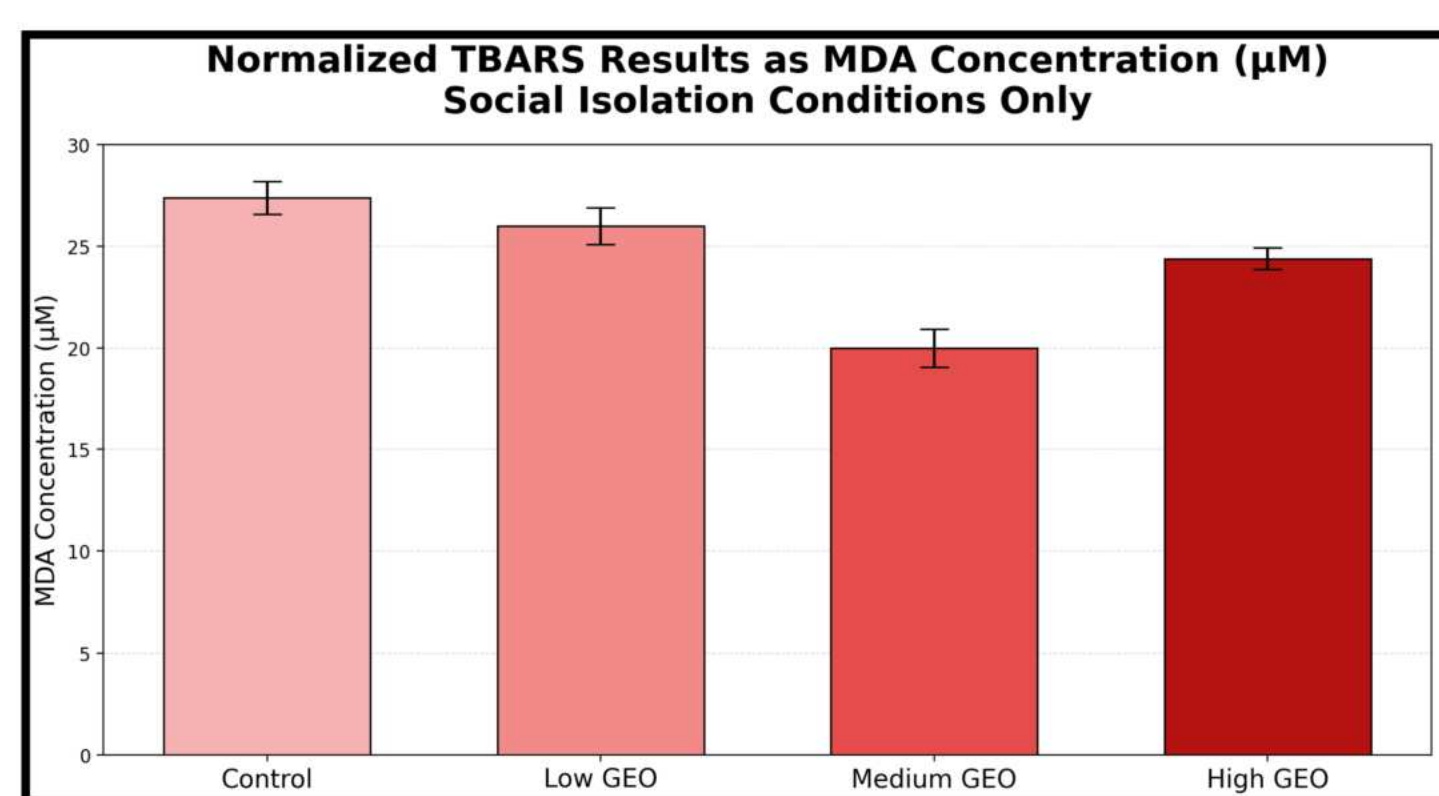


Figure 12: TBARS Isolation Experimental Results (Made by Researcher in Matplotlib)

- Baseline MDA in GH Control averaged 17.9 μM (range 16.5 to 20.0), setting the **low-stress reference level**.
- GH Medium GEO averaged 12.2 μM , a 5.7 μM decrease vs control, a **31.8% reduction**, the strongest effect in group housing.
- Dose pattern shows a clear minimum at medium dose: High GEO increased MDA by 1.5 μM vs Medium GEO, a **12.3% rebound from the best-performing dose**.
- Spread tightened at the low MDA doses: GH Medium GEO ranged 11.3 to 13.5, supporting consistent lowering across replicates.

- Baseline MDA in Isolation Control averaged 27.4 μM , a **53.1% increase** from Group Housing Control, showing strong isolation-linked oxidative burden.
- Isolation Medium GEO averaged 20.0 μM , a 7.4 μM decrease vs isolation control, a **27.0% reduction**, the strongest effect under isolation.
- Non-linear pattern is **strongest in isolation**: High GEO raised MDA by 4.4 μM vs Medium GEO, a 22.0% rebound from the best-performing dose.
- **Isolation** remained **higher** than group housing at every dose, including medium: Iso Medium GEO (20.0) stayed 7.8 μM above GH Medium GEO (12.2), a 63.9% difference, showing housing condition drives baseline oxidative damage even with GEO.

TBARS Group-Housing Experimental Results

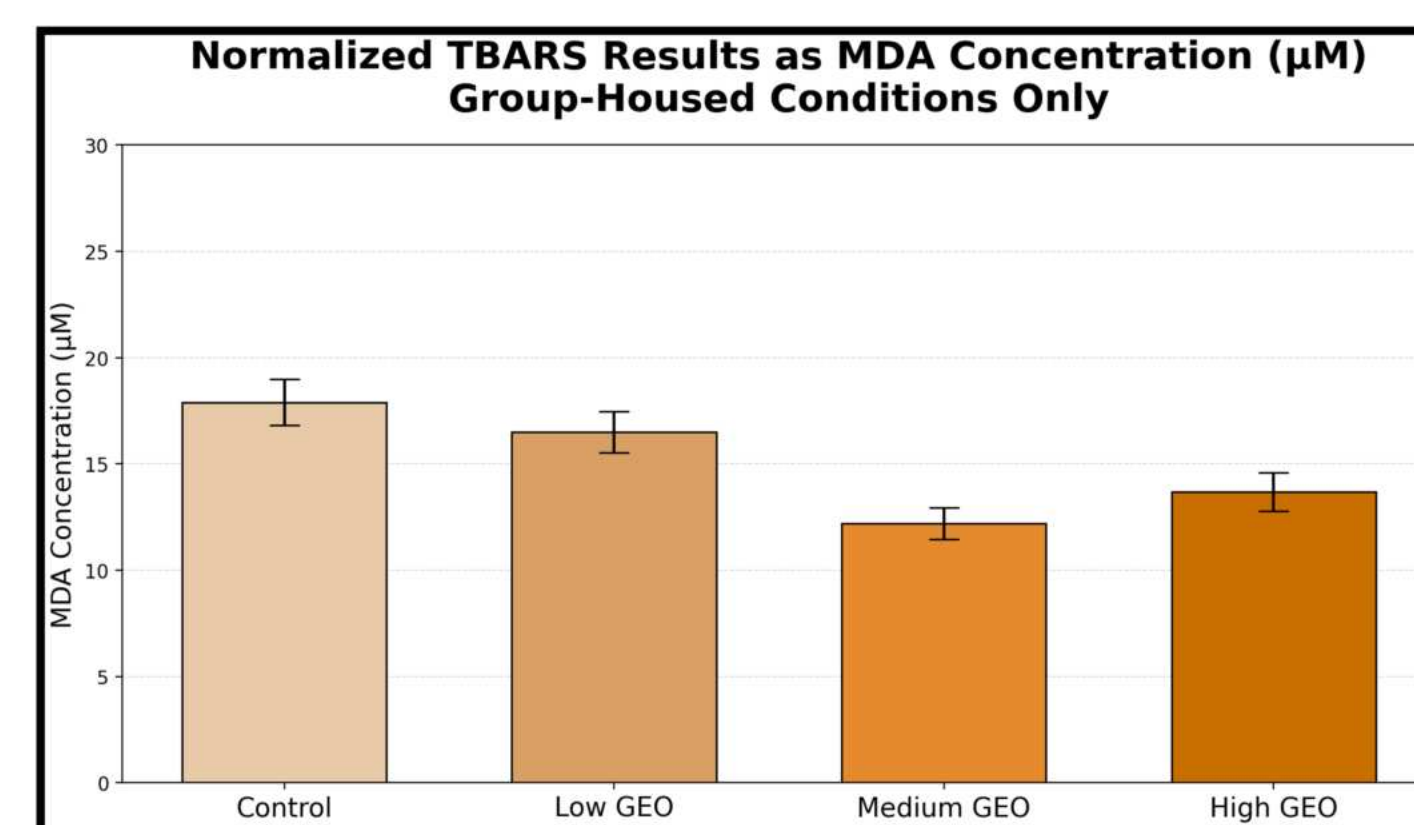


Figure 13: TBARS Group-Housing Experimental Results (Made by Researcher in Matplotlib)

Statistical Analysis

ANOVA F-statistic	P-value (ANOVA & Tukey)	Effect size
F(7,72) = 424.8	p = <0.001	eta² = 0.976
• Compares how far apart the MDA values are vs. how much MDA varies within each group. • The mean gaps between conditions are far larger than trial to trial noise .	• Measures how likely you would see an F-statistic this large if conditions had no impact on MDA. • Supports a real treatment effect, not a fluke from sample variation or error.	• Estimates the proportion of MDA variation explained by condition. • Nearly all MDA differences are explained by the experimental conditions .

Conclusion

Clear trend:

Isolation shifts baseline MDA **upward**, while **GEO** shifts MDA **downward** in both housing conditions, showing a strong, consistent change in lipid peroxidation across full design.

Dose pattern:

Medium GEO gives the **lowest MDA** in both conditions, while High GEO rebounds, supporting an optimal dose window and a non-linear dose response rather than "more dose equals more benefit."

Strength of evidence:

Group differences exceed trial-to-trial noise, so the effect is highly significant, meaning **condition explains almost all MDA variation**. This makes the impact strong and sets up direct next steps for mechanism validation with a second oxidative marker.

Medical Significance:

Shows that social isolation drives a large, measurable rise in lipid peroxidation and that ginger essential oil produces a strong reduction in MDA, **supporting a low-cost strategy** to quantify and mitigate stress-induced oxidative damage.

Key References

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